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Order Distribution Module

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Module Description: Requirements, business logic, algorithmic rules, and edge-case handling for the automated **Order Distribution System**.

1. Overview

The Order Distribution System automatically allocates orders from the **Claimable Pool** to available TeleSales Agents. The algorithm ensures fair and even distribution among eligible agents based on:

- **Agent Availability/Status** (Online vs. Offline)
- **Currently Pending Orders** count
- **Most Recently Assigned Order Timestamp**
- **Maximum Pending Order Limit** per agent (Hard limit: 10)

2. Key Concepts

Term	Definition
Claimable Pool	Global queue holding orders available for distribution to TeleSales Agents.
Currently Pending	Orders currently assigned to an agent that are awaiting processing/confirmation.
Online Agent	An agent actively logged in and marked available to receive new orders.
Offline Agent	An agent who is unavailable; strictly excluded from order distribution.
Maximum Pending Limit	Hard cap of 10 Currently Pending Orders per individual agent.
Recent Assignment	Timestamp indicating when an agent last received an order from the pool.
Least Recently Assigned	The eligible agent with the oldest assignment timestamp (longest wait time).

3. Eligibility & Algorithm Logic

Eligibility Matrix

An agent **MUST** meet all of the following criteria to enter the distribution pool:

1. Status is **Online**.
2. Currently Pending Orders count is **< 10**.
3. Account is active and marked available for assignment.

An agent is **EXCLUDED** from distribution if:

- Status is **Offline**.
- Currently Pending Orders count is **>= 10**.

Core Distribution Algorithm

When an order enters the Claimable Pool, the distribution engine executes the following steps:

1. **Filter**: Identify all agents matching the **Eligible Agent Criteria**.
2. **Sort**: Rank eligible agents by **Last Received Order Timestamp** in ascending order (*Least Recently Assigned first*).
3. **Assign**: Allocate the order to the top agent from the sorted list and update their `last_assigned_at` timestamp.

4. Distribution Examples & Scenarios

Scenario A: Initial Sequential Distribution (Equal Pending Counts)

All agents start Online with 0 pending orders.

Step	Action	Agent 1 (Pending)	Agent 2 (Pending)	Agent 3 (Pending)	Assigned To
0	Initial State	0	0	0	—
1	Order 1 arrives	1	0	0	Agent 1
2	Order 2 arrives	1	1	0	Agent 2
3	Order 3 arrives	1	1	1	Agent 3
4	Order 4 arrives	2	1	1	Agent 1 (<i>Longest wait time</i>)

Scenario B: Distribution Based on Recency (Unequal Pending Counts)

Priority is always governed by who has waited the longest since their last assigned order:

Step	State / Action	Agent 1	Agent 2	Agent 3	Assigned To / Reason
0	Baseline State	6 Pending (Recent)	4 Pending (Older)	2 Pending (Oldest)	—
1	Order 1 Arrives	6 Pending	4 Pending	3 Pending	Agent 3 (<i>Oldest timestamp</i>)
2	Order 2 Arrives	6 Pending	4 Pending	4 Pending	Agent 3 (<i>Still least recently assigned</i>)
3	Order 3 Arrives	6 Pending	4 Pending	4 Pending	Agent 2 or Agent 3 (<i>Evaluated by recency timestamp</i>)

Scenario C: Offline Agent Handling

Offline agents are automatically stripped from the distribution pool before evaluation.

State Matrix:

- **Agent 1:** 7 Pending (Online)
- **Agent 2:** 6 Pending (Online)
- **Agent 3:** 6 Pending (**Offline**)

Updated Result:

- **Agent 1:** 7 Pending (Online)
- **Agent 2:** **7 Pending** (Online)
- **Agent 3:** 6 Pending (**Offline**)

Scenario D: Maximum Limit & Completion Lifecycle

Hard Cap Enforcement (10 Pending)

Agent	Currently Pending	Status	Eligible for New Orders?
Agent 1	10	Online	 No (<i>Maximum limit reached</i>)
Agent 2	8	Online	 Yes
Agent 3	6	Online	 Yes

When a new order arrives, it is evaluated strictly between **Agent 2** and **Agent 3** based on assignment recency.

Re-entry on Order Completion

When **Agent 1** completes or confirms an order:

1. Pending order count drops: **10 → 9**.
2. Agent automatically re-enters the eligible distribution pool.

Event	Agent 1 Pending	Agent 2 Pending	Agent 3 Pending	Agent 1 Status
Initial State	10	8	7	 Ineligible
Agent 1 Completes 1 Order	9	8	7	 Eligible

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